MODULE – II

HTML 5 and Web Page Structuring HTML basics – elements, tags, attributes, document structure. Text formatting and hyperlinks. Lists, tables, anchors, forms and input elements. HTML5 semantic tags – header, footer, nav, article, section. Embedding media – audio, video, iframes. Canvas and SVG basics. Responsive design with HTML meta viewport. Accessibility standards – ARIA roles, alt text, best practices.

INTRODUCTION TO HTML:

- HTML stands for **H**yper **T**ext **M**arkup **L**anguage.
- HTML is a method of describing the format of web documents
- It is used to display the document in the web browsers.
- HTML was developed by Tim Berners-Lee. HTML standards are created by a group of interested organizations called W3C (World Wide Web consortium).

HTML Tags:

- In HTML, formatting is specified by using tags.
- A tag is a format name surrounded by angle brackets.
- End tags which switch a format off also contain a forward slash.

Points to be remembered for HTML tags:

- They are not case sensitive i.e., <head>, <HEAD> and <Head> is equivalent.
- If a browser does not understand a tag it will usually ignore it.
- White spaces, tabs and newlines are ignored by the browser.

STRUCTURE OF A HTML DOCUMENT:

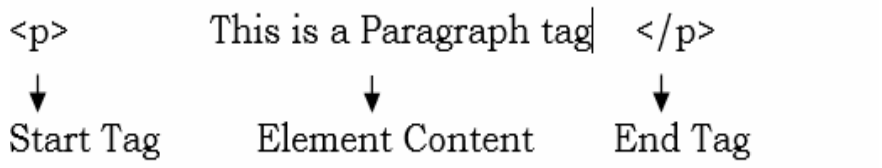
- HTML document consists of 2 sections.
  1. Head Section
  2. Body Section
- The basic document is shown below.

```
<html>
    <head>
        <! -- Head Section -->
    </head>
    <body>
```

```
                <! -- Body Section -->
            </body>
        </html>
```

HTML ELEMENTS:

An HTML element is everything from the start tag to the end tag



HTML ATTRIBUTES:

HTML Elements can have Attributes. Attribute provide additional information about an element and are always specified in the start tag.

Syntax: <tagname attributename="value" > Content </tag>

Sample.html

```
<html>
    <head>
        <title> Basic HTML document </title>
    </head>
    <body>
        <h1> Welcome to the world of Web Technologies</h1>
        <p> A sample HTML program </p>
    </body>
</html>
```

Output:



BASIC HTML TAGS:

1. **<html> :**
   - The <html> tag tells the browser that this is an HTML document.
   - The <html> tag represents the root of an HTML document.
   - The <html> tag is the container for all other HTML elements.
2. **<title>:**
   - defines a title in the browser toolbar
   - provides a title for the page when it is added to favorites
   - displays a title for the page in search-engine results
3. **<body>:**
   - The <body> tag defines the document's body.
   - The <body> element contains all the contents of an HTML document, such as text, hyperlinks, images, tables, lists, etc.

Attribute	Value	Meaning
background	URL	Specifies a background image for a document
Bgcolor	Color yellow rgb(255,255,0) #ffff00	Specifies the background color of a document
Text	Color	Specifies the color of the text in a document
Vlink	Color	Specifies the visited hyper link color
Link	Color	Specifies the hyper link color
Alink	Color	Specifies the active hyper link color

4. **<!- - - - > Comment Tag:**
   - The comment tag is used to insert comments in the source code. Comments are not displayed in the browsers.
5. **Heading Tags:**
   - There are 6 heading tags.
   - The <h1> to <h6> tags are used to define HTML headings.

- <h1> defines the most important heading. <h6> defines the least important heading.

Attribute	Value	Meaning
Align	Left Right Center	Specifies the alignment of a heading

Example: Headings.html

```
<html>
    <head>
        <title>Heading Tage</title>
    </head>
    <body bgcolor=yellow text=blue>
        <! - - This is a Comment - - >
        <h1 align="left">This is Heading 1</h1>
        <h2 align="center">This is Heading 2</h2>
        <h3 align="right">This is Heading 3</h3>
        <h4>This is Heading 4</h4>
        <h5>This is Heading 5</h5>
        <h6 align="right">This is Heading 6</h6>
        </body>
</html>
```

Output:



6. **<p>: paragraph Tag**

- Browser automatically add some space before and after each <p> element

Attribute	Value	Meaning
Align	Left, Right, Center Justify	Specifies the alignment of text within a paragraph

7. **<font>:**

The <font> tag specifies the font face, font size, and color of text.

Attribute	Value	Meaning
Color	rgb(x,x,x) #xxxxxx colorname	Specifies the color of text
Face	font_family	Specifies the font of text
Size	Number	Specifies size of text

2. ** :**

- The tag defines a division or a section in an HTML document.
- The tag is used to group block-elements to format them with CSS.

Attribute	Value	Meaning
Align	Left, Right, Center Justify	Specifies the alignment of a heading

3. **
:**

- The
 tag inserts a single line break.
- The
 tag is an empty tag which means that it has no end tag.

It is used for Scrolling images and text in the web page

Attribute	Value	Meaning
behavior	Scroll, slide alternate	Defines the type of scrolling.
bgcolor	rgb(x,x,x) #xxxxxx colorname	*Deprecated*-Defines the direction of scrolling the content.
direction	Up, down, left, right	Defines the direction of scrolling the content.
Loop	Number	Specifies how many times to loop. The default value is INFINITE, which means that the marquee loops endlessly.
scrolldelay	Seconds	Defines how long to delay between each jump.
scrollamount	Number	Defines how how far to jump.

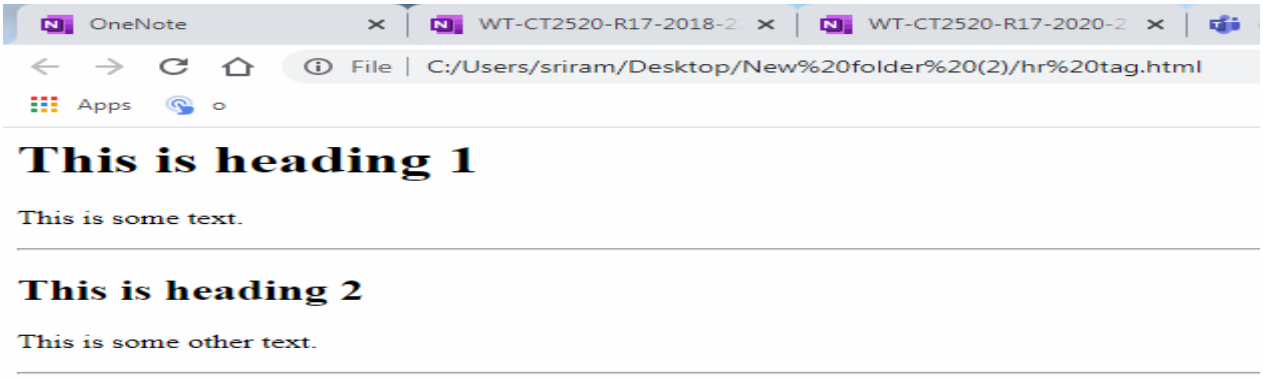
14. HTML Horizontal Rules

The <hr> tag defines a thematic break in an HTML page, and is most often displayed as a horizontal rule.

The <hr> element is used to separate content (or define a change) in an HTML page:

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.0 Transitional//EN">
<HTML>
 <HEAD> </HEAD>
 <BODY>
<h1>This is heading 1</h1><p>This is some text.</p>
<hr>
<h2>This is heading 2</h2><p>This is some other text.</p>
<hr>
 </BODY>
</HTML>
```

Output:



➢ **SECTION ELEMENTS:**

Section Elements are used to define the scope of header, footers and various other sections of an HTML document. Different section elemets are used to define different pas of the document. The following are the list of section elements:

- BODY
- SECTION

- NAV
- ARTICLE
- ASIDE
- HEADER
- FOOTER
- ADDRESS

BODY: The body element contains all the contents of an HTML document, such as headings, paragraphs, images, hyperlinks, tables, lists, etc. The BODY element is placed after the closing tag of HEAD element in HTML element.

```
<HTML>
        <head>
        </head>
    <body>
        <p>This in Paragraph Tag</p>
    </body>
</HTML>
```

SECTION: Section element is a generic section of HTML document or application, and should only be used if there isn't a more specific element to represent it.

```
 <section>
 <h2>WWF History</h2>
    <p>The World Wide Fund for Nature (WWF) is an international organization.</p>
</section>
<section>
 <h2>WWF's Symbol</h2>
   <p>The Panda has become the symbol of WWF. </p>
</section>
```

NAV: Nav element is newly introduced in HTML5 and is used to navigate from one HTMl page to other. Notice that NOT all links of a document should be inside a <nav> element. The <nav> element is intended only for major block of navigat

```
<nav>
 <a href="/article.html/">HTML</a> |
 <a href="/article.css/">CSS</a> |
 <a href="/ article.js/">JavaScript</a> |
 <a href="/ article.python/">Python</a>
```

```
</nav>
```

ARTILCE: Article element is newly introduced in HTML5. The<article> tag specifies independent, self-contained content. An article should make sense on its own and it should be possible to distribute it independently from the rest of the site.

```
<article>
  <h2>Google Chrome</h2>
   <p>Google Chrome is a web browser developed by Google, released in 2008.
Chrome is the world's most popular web browser today!</p>
</article>

<article>
  <h2>Mozilla Firefox</h2>
  <p>Mozilla Firefox is an open-source web browser developed by Mozilla.</p>
</article>
```

ASIDE: ASIDE element is introduced in HTML5. The <aside> tag defines some content aside from the content it is placed in. The aside content should be indirectly related to the surrounding content.

- The <aside> content is often placed as a sidebar in a document.

```
<aside>

  <h4>FSAD</h4>

   <p>Full Stack Application Development course contains both Front-end and
back-end design</p>

</aside>
```

Header: The <header> element represents a container for introductory content or a set of navigational links.

A <header> element typically contains:

- one or more heading elements (<h1> - <h6>)
- logo or icon
- authorship information

Note: You can have several <header> elements in one HTML document. However, <header> cannot be placed within a <footer>, <address> or another <header> element.

```
<article>
  <header>
```

```
  <h1>A heading here</h1>
  <p>Posted by John Doe</p>
  <p>Some additional information here</p>
</header>
```

Footer: Footer is element is introduced in HTML5. A <footer> element typically contains information such as authorship information, copyright information, contact information, sitemap, back to top links, related documents.

```
<footer>
  <p>Author: FSAD <br>
  <a href="mailto:mani@gecgudlavallerumic.in">nagamani</a></p>
</footer>
```

Address: Address is introduced in HTML5. The <address> tag defines the contact information for the author/owner of a document or an article. The contact information can be an email address, URL, physical address, phone number, social media handle, etc.

```
<Address>
        Asst.Professor
        CSE,
        SRGEC, Gudlavalleru
</address>
```

- **TEXT FORMATTING TAGS:**

The following HTML tags are used for format the appearance of the text on your web page.

(a). Headings – <h1> to <h6>

(b). Bold -   or

The text in between the tags will be displayed in bold

(c). Italic -

Renders the text in italics i.e displays the text at a slight angle.

(d). Underline - <u> </u>

Underlines the text written in between the tags

(e). Strike out - <strike> </strike>

Defines strike through text, puts a line right through the center of the text, crossing it out.

(f).Preformatted text - <pre> </pre>

Text in <pre> element is displayed in fixed width font, and it preserves both spaces and line breaks.

(g). Typewriter Text - <tt> </tt>

The text appears to have been typed by a type writer

(h). <big> </big> - Defines bigger text

(i). <small> </small> - Defines smaller text

(j). - Defines a subscript text. Subscript that appears half a character below the baseline.

(k). - Defines a superscript text. Superscript that appears half a character above the baseline.

(l). - It align the text to the center of the page

Example: TextFormattingTags.html

```
<html>
  <body>
    <h1>This is Heading 1</h1>
    <b>This text is in bold</b><br>
    <i>This text is in Italics</i><br>
    <u>This text is in Underlined</u><br>
    <strike>This text is Striked</strike><br>
    <em>This text is Emphasized</em><br>
    <tt>This text is Type Writer Text</tt><br>
    <big>This text is Bigger</big><br>
    <small>This text is Smaller</small><br>
    H<sub>2</sub>O<br>
    (a+b)<sup>2</sup>=a<sup>2</sup>+2ab+b<sup>2</sup><br>
    <center>This Text is aligned to Center</center><br>
</body>
</html>
```

Output:



- **LINKS:**

1. **<a>: Anchor Tag**

- The <a> tag defines a hyperlink, which is used to link from one page to another.
- The most important attribute of the <a> element is the href attribute, which indicates the link's destination.
- By default, links will appear as follows in all browsers:
  - An unvisited link is underlined and blue
  - A visited link is underlined and purple
  - An active link is underlined and red

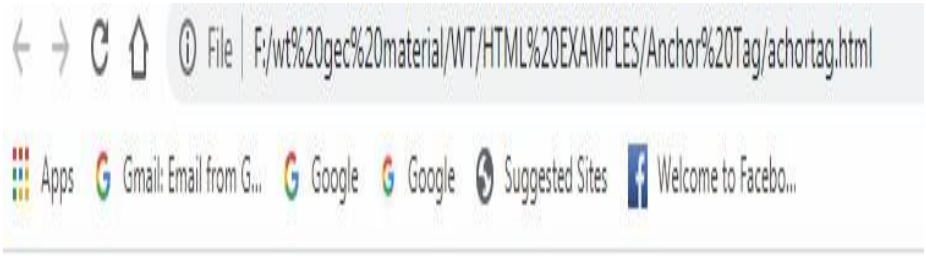
Attribute	Value	Meaning
Href	URL	Specifies the destination of the link
Target	_blank _self _parent _top framename	Specifies where to open the linked document

Example: Link.html

```
<HTML>
<HEAD>
```

```
 <TITLE> Anchor Tag Example </TITLE>
</HEAD>
<BODY>
    <a href='https://www.facebook.com' target='_blank'>Click Here to Visit Facebook</a><br>
    <a href='https://www.gmail.com' target='_top'>Click Here to Gmail</a>
</BODY>
</HTML>
```

Output:



Absolute URLs vs. Relative URLs

Both examples above are using an absolute URL (a full web address) in the href attribute.

A local link (a link to a page within the same website) is specified with a relative URL (without the "https://www" part):

Example:

```
<h2>Absolute URLs</h2>
<p><a href="https://www.w3.org/">W3C</a></p>
<p><a href="https://www.google.com/">Google</a></p>

<h2>RelativeURLs</h2>
<p><a href="html_images.asp">HTMLImages</a></p>
<p><a href="/css/default.asp">CSS Tutorial</a></p>
```

To use an image as a link, just put the <img> tag inside the <a> tag:

Example:

```
<a href="default.asp">
<img src="smiley.gif" alt="HTML tutorial" style="width:42px; height:42px;">
</a>
```

Use mailto: inside the href attribute to create a link that opens the user's email program

Send email

4. **<link>:**

- The <link> tag defines a link between a document and an external resource.
- The <link> tag is used to link to external style sheets.

Attribute	Value	Meaning
Href	URL	Specifies the location of the linked document
Target	_blank _self _parent _top framename	Specifies where the linked document is to be loaded
Rel	Stylesheet	Specifies the relationship between current document and the linked document

➢ WORKING WITH LISTS

- Lists are used to collect a group of items.
- There are 3 types of Lists in HTML
    1. Ordered List
    2. Unordered List
    3. Definition List

1. **ORDERED LIST:**

- These are those in which the items are arranged in some specific order.
- This list can be numerical or alphabetic.
- ** tag:** The <ol> tag defines an ordered list.

Attributes:

Name	Value	Meaning
type	1,2,3,4.. A,B,C,D a,b,c,d. I,II,III,IV i,ii,iii,iv	Specifies the kind of marker to use in the list

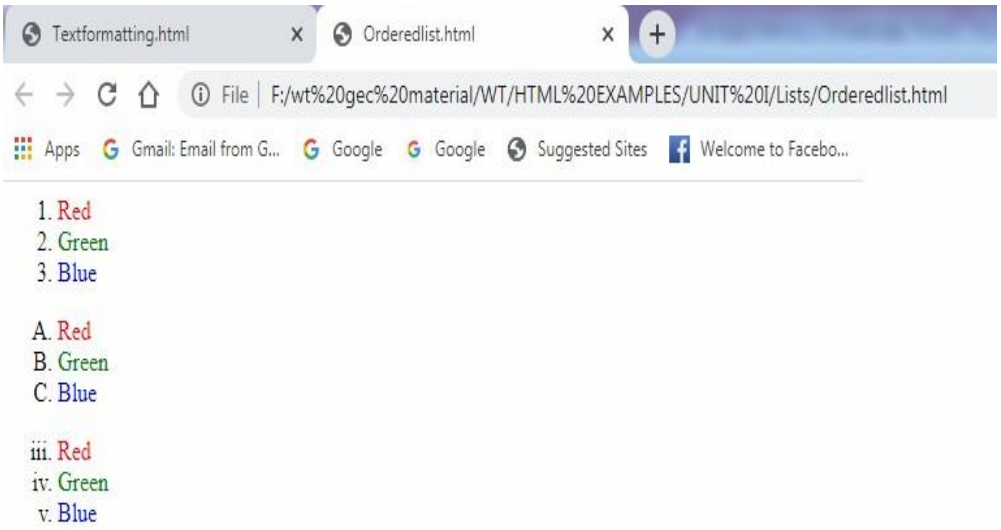
start	number	Specifies the start value of an ordered list
reversed	reversed	Specifies that the list order should be descending

- ** tag:** defines a list item.
- **Example:**

```
<html>
<ol>
        <li><font color='red'>Red</font></li>
        <li><font color='green'>Green</font></li>
        <li><font color='blue'>Blue</font></li>
</ol>
```

```
<ol type="A">
        <li><font color='red'>Red</font></li>
        <li><font color='green'>Green</font></li>
        <li><font color='blue'>Blue</font></li>
</ol>
<ol start=3 type="i">
        <li><font color='red'>Red</font></li>
        <li><font color='green'>Green</font></li>
        <li><font color='blue'>Blue</font></li>
</ol>
</body>
</html>
```

Output:



2. **UNORDERED LIST:**

- The Unordered lists are those in which the items are not arranged in any order.
- This defines a Bulleted List.
- ** tag**: defines an unordered (bulleted) list.

  Attributes:

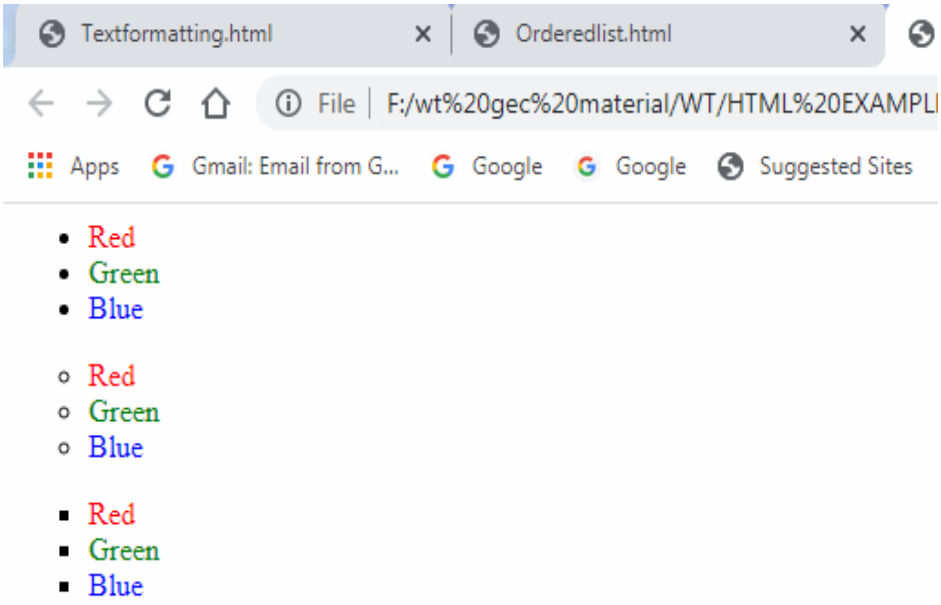
Name	Value	Meaning
Type	• Disc ▪ Square ○ Circle	Specifies the kind of marker to use in the list

- ** tag:** defines a list item.
- **Example:**

```
<html>
<body>
        <ul>
                <li>Red</li>  <li>Black</li>
                <li>White</li>
        </ul>
        <ul type="circle">
                <li>Red</li>            <li>Black</li>

                <li>White</li>
        </ul>
        <ul type="square">
                <li>Red</li>  <li>Black</li>
                <li>White</li>
        </ul>
</body>
</html>
```

Output:



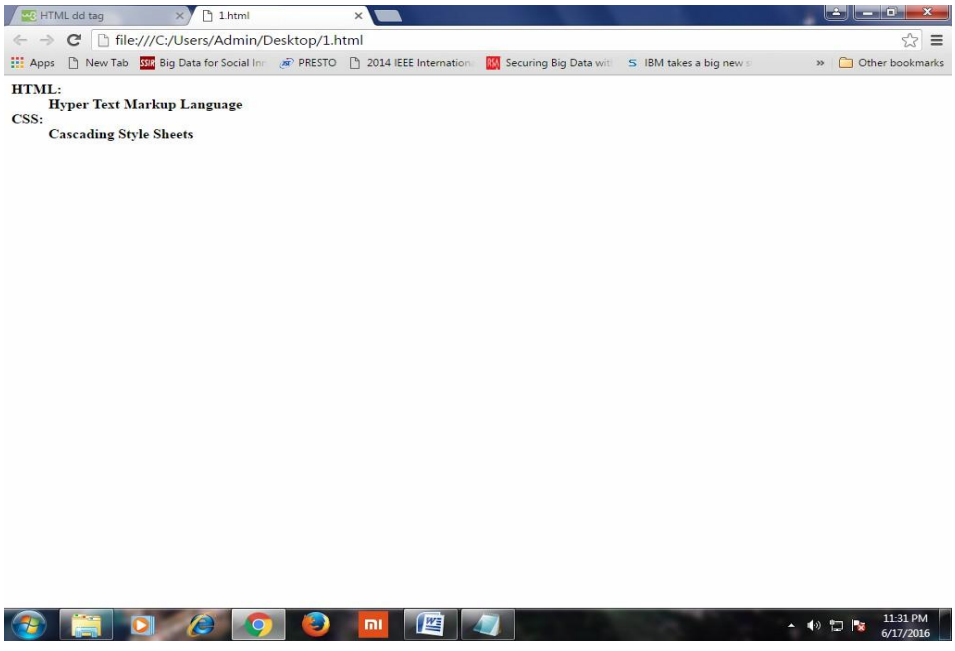
3. **DEFINITION LIST**

- These are lists of items that have 2 parts, a term to be defined and the definition.
- This create lists similar to a dictionary.
  - **<dl> tag:** defines a definition list. It is used in conjunction with <dt> and <dd>
  - **<dt> tag:** defines a term/name in a definition list.
  - **<dd> tag:** used to describe a term/name in a definition list.
- **Example:**

```
<html>
    <body>
        <dl>
            <dt>HTML:</dt>
            <dd>Hyper Text Markup Language</dd>

            <dt>CSS:</dt>
            <dd>Cascading Style Sheets</dd>
        </dl>
</body>
</html>
```

Output:



- **TABLES:**
  - For Systematic arrangement of information we often require Tabular Structure.
  - The biggest advantage of using tables on the web page is that the information gets arranged systematically.
  - The <table> tag defines an HTML table.
  - An HTML table consists of the <table> element and one or more <tr>, <th>, and<td> elements.
  - The <tr> element defines a table row, the <th> element defines a table header, and the <td> element defines a table cell.
  - An HTML table has two kinds of cells:
    - Header cells - contains header information (created with the <th> element)
    - Standard cells - contains data (created with the <td> element)
  - The text in <th> elements are bold and centered by default.
  - The text in <td> elements are regular and left-aligned by default.

- **Attributes of <table> tag:**

Name	Value	Meaning
align	Left Right Center	Specifies the alignment of a table according to surrounding text
bgcolor	rgb(x,x,x) #xxxxxx colorname	Specifies the background color for a table
border	0 1	Specifies whether or not the table is being used for layout purposes
cellpadding	Pixels	Specifies the space between the cell wall and the cell content
cellspacing	Pixels	Specifies the space between cells
Width	Pixels %	Specifies the width of a table

- **Attributes of <tr> tag:**

Name	Value	Meaning
align	Left Right Center Justify	Aligns the content in a table row
bgcolor	rgb(x,x,x) #xxxxxx Colorname	Specifies a background color for a table row
valign	Top middle bottom baseline	Vertical aligns the content in a table row

- **Attributes of <th> and <td> tags:**

Name	Value	Meaning
Align	Left, Right Center Justify	Aligns the content in a cell
bgcolor	rgb(x,x,x) #xxxxxx Colorname	Aligns the content in a cell
rowspan	Number	Specifies the number of rows a cell should span
colspan	Number	Specifies the number of rows a cell should span
valign	Top, middle	Vertical aligns the content in a cell

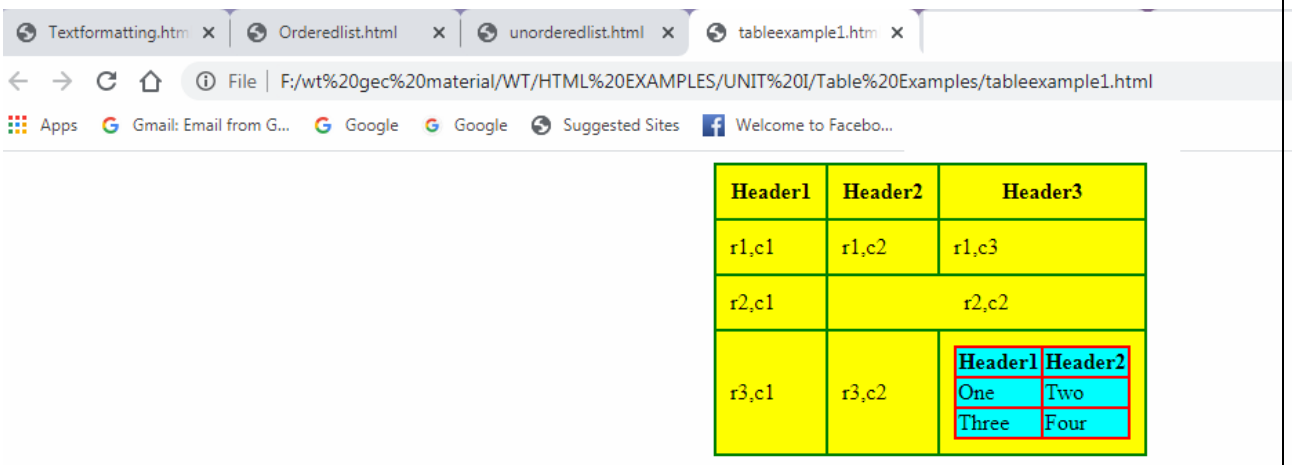
	bottom baseline	
Width	Pixels %	Specifies the width of a cell

- **Example:**

```
<html>
<body>
<table bgcolor="yellow" border="1" cellspacing="0" cellpadding="10" bordercolor="green" align="center">
<tr>
        <th>Header1</th>
        <th>Header2</th>
        <th>Header3</th>
</tr>
<tr>
        <td>r1,c1</td>
        <td>r1,c2</td>
        <td>r1,c3</td>
</tr>
<tr>
        <td>r2,c1</td>
        <td colspan='2' align='center'>r2,c2</td>
</tr>
<tr>
        <td>r3,c1</td>
        <td>r3,c2</td>
        <td>
                <table border="1" bgcolor="cyan" cellspacing="0" bordercolor="red">
                <tr>
                        <th>Header1</th>
                        <th>Header2</th>
                </tr>
                <tr>
                        <td>One</td>
                        <td>Two</td>
                </tr>
                <tr>
                        <td>Three</td>
                        <td>Four</td>
                </tr>
                </table>
        </td>
</tr>
</table>
</body>
```

</html>

Output:



➢ FORMS:

- Form is a typical layout on the web page by which an user can interact with the web page.
- The <form> tag is used to create an HTML form for user input.
- The <form> element can contain one or more of the following form elements:
  - <input> <textarea> <select> <option> <label>
- **Attributes of <form> tag:**

Name	Value	Meaning
action	*URL*	Specifies where to send the form-data when a form is submitted
method	get post	Specifies the HTTP method to use when sending form-data
name	*text*	Specifies the name of a form
target	_blank _self _parent _top	Specifies where to display the response that is received after submitting the form

Form Elements:

- **<input>:**
  - The <input> tag specifies an input field where the user can enter data.

- <input> elements are used within a <form> element to declare input controls that allow users to input data.
- An input field can vary in many ways, depending on the type attribute.
- **Attributes of <input > tag:**

Name	Value	Meaning
Type	**Button, checkbox** **dat**e, **file**, hidden image, month, number **password**, **radio**, **reset** **submit, text**	Specifies the type <input> element to display
Name	Text	Specifies the name of an <input> element
Checked	Checked	Specifies that an <input> element should be pre-selected when the page loads (for type="checkbox" or type="radio")
Value	*Text*	Specifies the value of an <input> element

- **<textarea>:**
  - The <textarea> tag defines a multi-line text input control.
  - A text area can hold an unlimited number of characters, and the text renders in a fixed-width font (usually Courier).
  - The size of a text area can be specified by the cols and rows attributes
  - **Attributes of <textarea > tag:**

Name	Value	Meaning
Name	Text	Specifies a name for a text area
Rows	number	Specifies the visible number of lines in a text area
Cols	number	Specifies the visible width of a text area

- **<select>:**
  - The <select> element is used to create a drop-down list.
  - The <option> tags inside the <select> element define the available options in the list.
  - **Attributes of <select > tag:**

Name	Value	Meaning
Name	name	Defines a name for the drop-down list

Multiple	multiple	Specifies that multiple options can be selected at once
Size	number	Defines the number of visible options in a drop-down list

Example:

```
<select name="cars">
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="fiat">Fiat</option>
  <option value="audi">Audi</option>
</select>
```

Visible Values:

Use the size attribute to specify the number of visible values:

Example

```
<label for="cars">Choose a car:</label>
<select id="cars" name="cars" size="3">
  <option value="1">Volvo</option>
  <option value="2">Saab</option>
  <option value="3">Fiat</option>
  <option value="4">Audi</option>
</select>
```

Allow Multiple Selections:

Use the multiple attribute to allow the user to select more than one value:

Example

```
<label for="cars">Choose a car:</label>
<select id="cars" name="cars" size="4" multiple>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="fiat">Fiat</option>
  <option value="audi">Audi</option>
</select>
```

- **<label>:** The <label> tag defines a label for an <input> element.

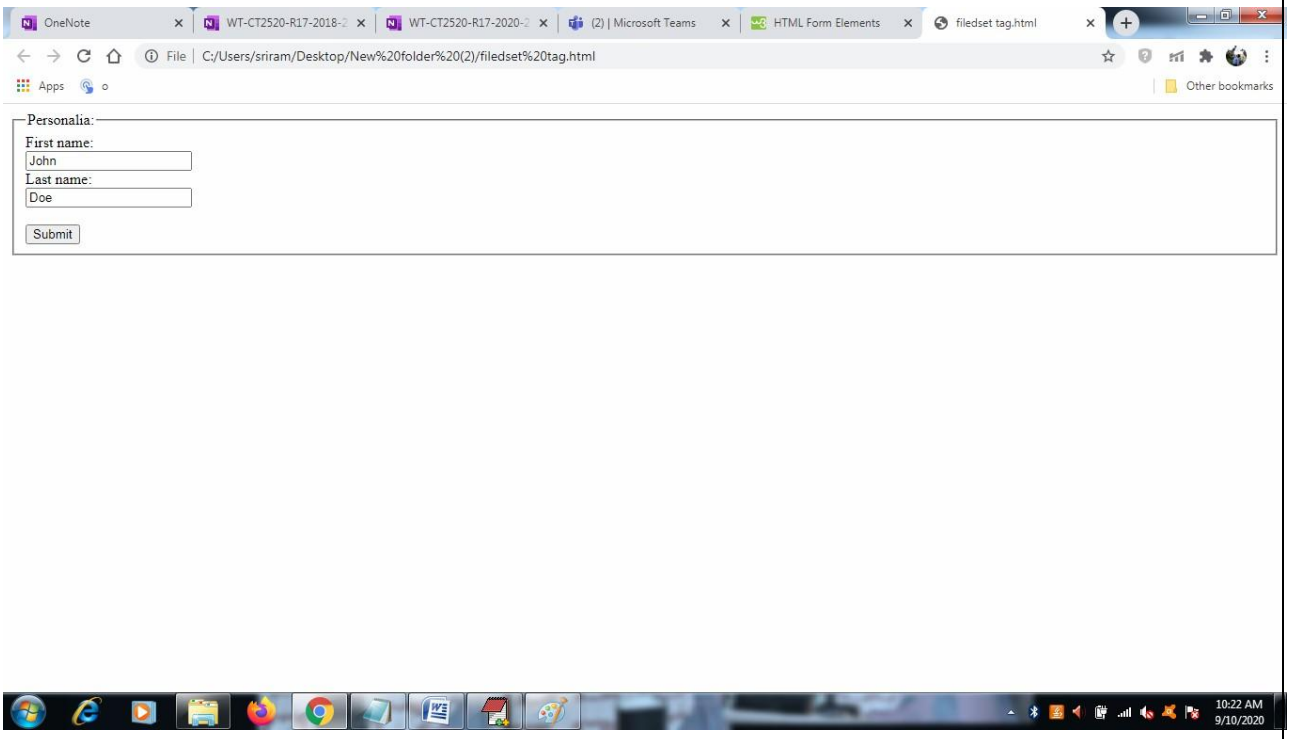
- The <label> element is useful for screen-reader users, because the screen-reader will read out loud the label when the user focus on the input element.
- **The <fieldset> and <legend> Elements**

  The <fieldset> element is used to group related data in a form.

  The <legend> element defines a caption for the <fieldset> element.

  Example

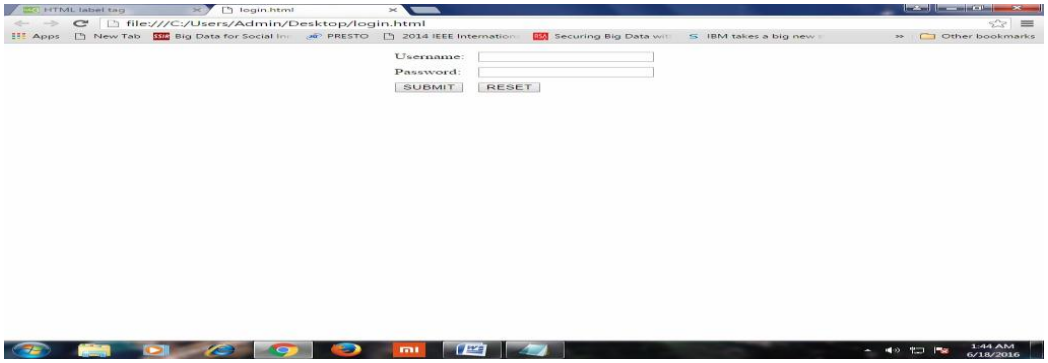
```
<form action="/action_page.php">
  <fieldset>
    <legend>Personalia:</legend>
    <label for="fname">First name:</label><br>
    <input type="text" id="fname" name="fname" value="John"><br>
    <label for="lname">Last name:</label><br>
    <input type="text" id="lname" name="lname" value="Doe"><br><br>
    <input type="submit" value="Submit">
  </fieldset>
</form>
```



- **Example:** Login.html

```
<html>
<body>
<form name="f1" method="post" action="">
<table align="center" cellspacing="10">
<tr>
<td><label> Username: </label> </td>
<td><input type="text" name="t1"></td>
</tr>
<tr>
<td><label> Password: </label></td>
<td><input type="password" name="t2"></td>
</tr>
<tr>
<td><input type="submit" value="SUBMIT" ></td>
<td><input type="reset" value="RESET" ></td>
</form>
</body>
</html>
```

Output:



➢ FRAMES:

- HTML Frames divide a browser window into several pieces or panes, each pane containing a separate HTML page.
- Each portion is called as a ***Frame***.
- A Collection of Frames in the browser window is known as a ***Frameset***.
- HTML Frames allow authors to present documents in multiple views, which may be independent windows or sub windows.
- One of the Key advantages that frames offer is that you can load and reload single frames without having to reload the entire contents of the browser window.
- **<frameset>:**

- The <frameset> tag defines a frameset.
- The <frameset> element holds one or more <frame> elements. Each <frame> element can hold a separate document.
- The <frameset> element specifies how many columns or rows there will be in the frameset, and how much percentage/pixels of space will occupy each of them.
- **Attributes of <frameset> tag:**

Name	Value	Meaning
cols	Pixels, % *	Specifies the number and size of columns in a frameset
rows	Pixels, % *	Specifies the number and size of rows in a frameset

- **<frame>:**
  - The <frame> tag defines one particular window (frame) within a <frameset>.
  - Each <frame> in a <frameset> can have different attributes, such as border, scrolling, the ability to resize, etc.
  - **Attributes of <frame> tag:**

Name	Value	Meaning
Src	URL	Specifies the URL of the document to show in a frame
frameborder	0 1	Specifies whether or not to display a border around a frame
name	*text*	Specifies the name of a frame
noresize	noresize	Specifies that a frame is not resizable
scrolling	yes no auto	Specifies whether or not to display scrollbars in a frame

- **Example:**

Frame.html:

```
<html>
<frameset cols="25%,*">
  <frame   src="links.html" name="Frame1" >
   <frame name="Frame2">
</frameset>
</html>
```

Links.html page it is included in the frame1

Links.html:

```
<html>
<body link=green alink=yellow vlink=orange>
<h1>
<center>
<a href=one.html  target=Frame2>Car 1</a><br>
<a href=two.html  target=Frame2>Car 2</a><br>
<a href=three.html target=Frame2>Car 3</a><br>
</center>
</h1>
</body>
</html>
```

Now following one.html, two.html and three.html are included in the frame2

One.html:

```
<HTML>
 <BODY>
  <img src="one.jpg">
 </BODY>
</HTML>
```

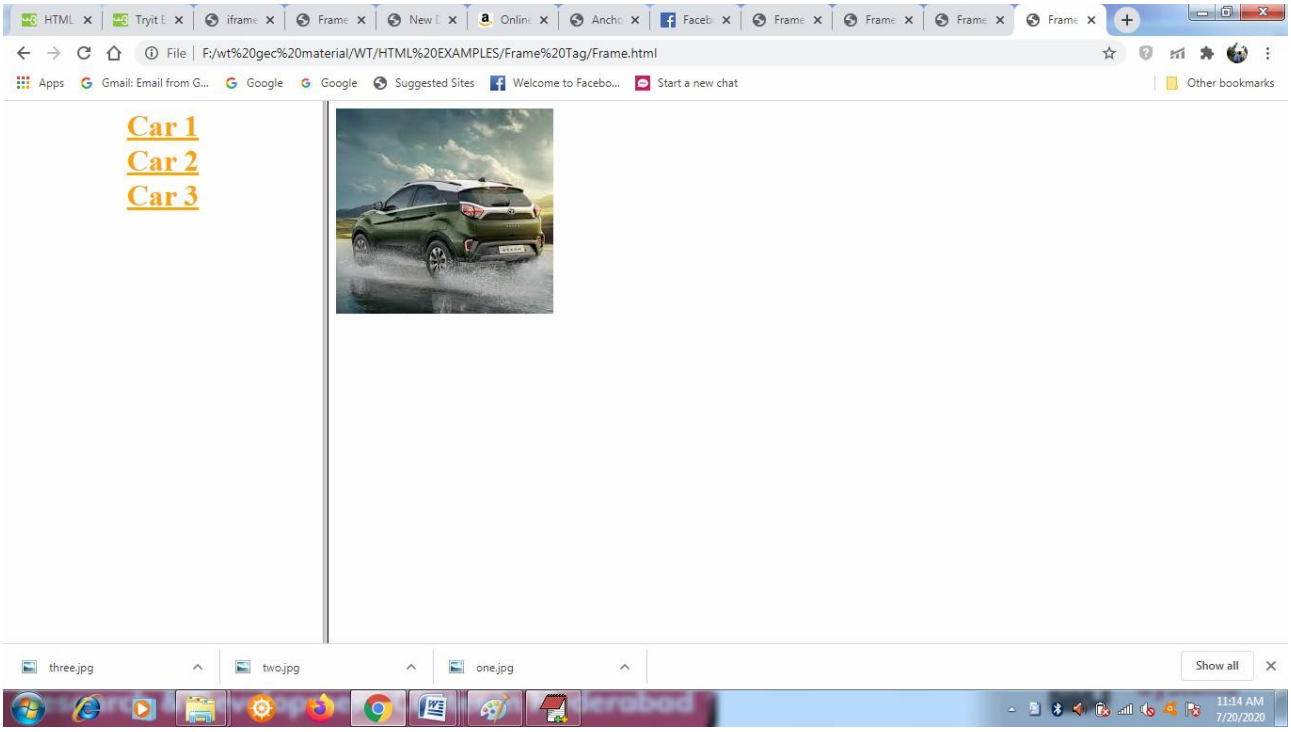
Two.hmtl:

```
<HTML>
 <BODY>
  <img src="two.jpg">
 </BODY>
</HTML>
```

Three.html:

```
<HTML>
 <BODY>
  <img src="three.jpg">
 </BODY>
</HTML>
```

Output:



Inline Frames - The <iframe> Element:

You can define an inline frame with the <iframe> tag. The <iframe> tag is not used within a <frameset> tag. Instead, it appears anywhere in your document. The <iframe> tag defines a rectangular region within the document in which the browser displays a separate document, including scrollbars and borders.

Use the *src* attribute with <iframe> to specify the URL of the document that occupies the inline frame.

All of the other, optional attributes for the <iframe> tag, including *name, class, frameborder, id, longdesc, marginheight, marginwidth, name, scrolling, style, and title* behave exactly like the corresponding attributes for the <frame> tag.

Following is the example to show how to use the <iframe>. This tag is used along with <body> tag:

iframe.html:

```
<html>
<body>
<h1>The iframe element + CSS</h1>
<p>An iframe with default borders:</p>
<iframe src="one.html" width="100%" height="100">
</iframe>
<p>An iframe with a thin black border:</p>
<iframe src="two.html" width="100%" height="100" style="border:1px solid black;">
</iframe>
<p>An iframe with no borders:</p>
<iframe src="three.html" width="100%" height="100" style="border:none;">
</iframe>
</body>
</html>
```

Another three documents which are embedded in above document are one.html, two.html and three.html are following

One.html:

```
<HTML>
 <BODY>
  <img src="one.jpg">
 </BODY>
</HTML>
```

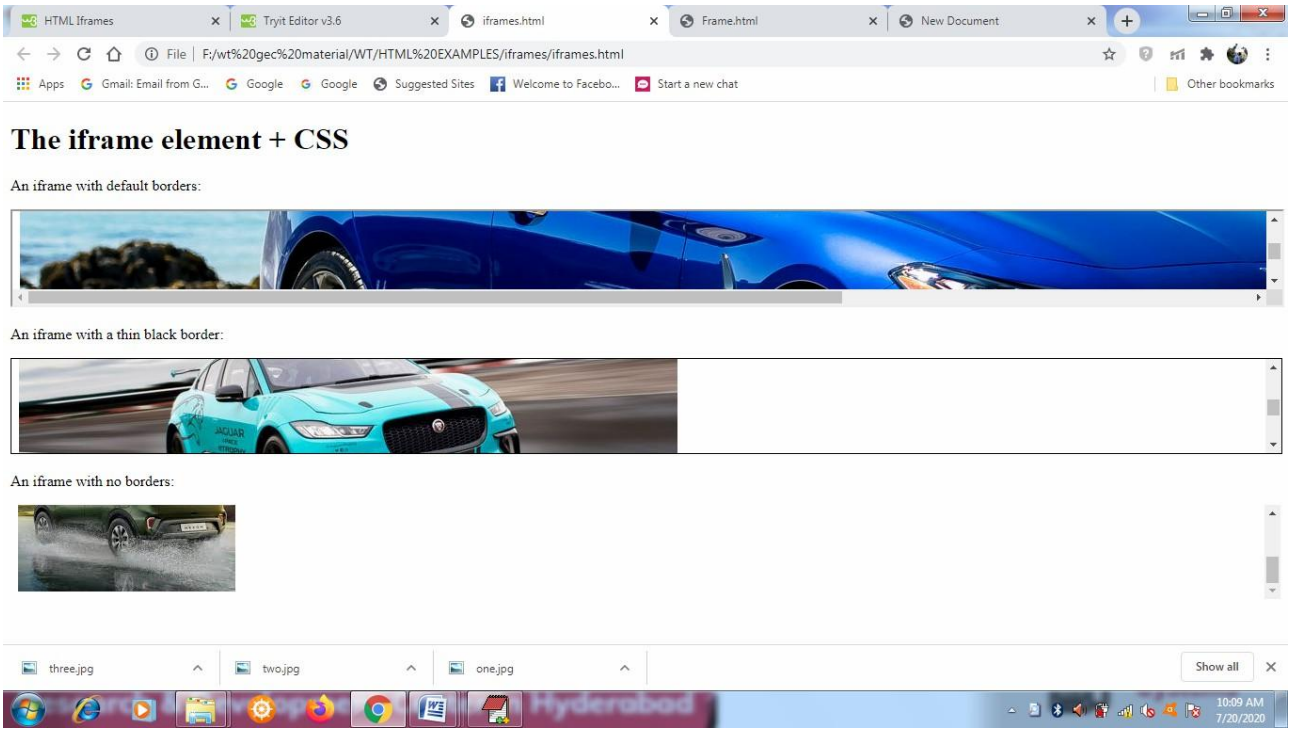
Two.hmtl:

```
<HTML>
 <BODY>
  <img src="two.jpg">
 </BODY>
</HTML>
```

Three.html:

```
<HTML>
```

```
<BODY>
 <img src="one.jpg">
</BODY>
</HTML>
```



➢ IMAGES:

- Images increase the visual appearance of web pages and make your web pages more attractive.
- The <img> tag defines an image in an HTML page.
- The <img> tag has two required attributes: src and alt.
- **Attributes of <img> tag:**

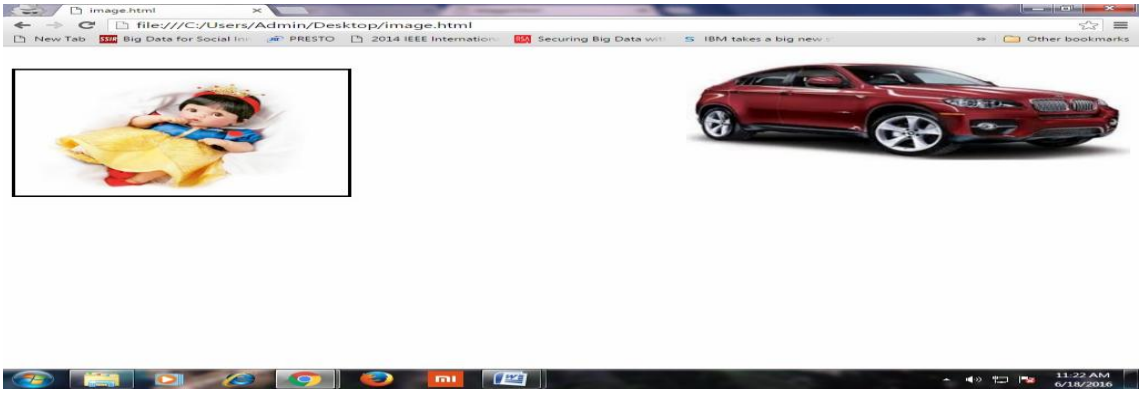
Name	Value	Meaning
src	URL	Specifies the URL of an image
align	Top, bottom middle left, right	Specifies the alignment of an image according to surrounding elements
alt	Text	Specifies an alternate text for an image
border	Pixels	Specifies the width of the border around an image

width	Pixels	Specifies the width of an image
height	Pixels	Specifies the height of an image
hspace	Pixels	Specifies the whitespace on left and right side of an image
vspace	Pixels	Specifies the whitespace on top and bottom of an image
ismap	Ismap	Specifies an image as a server-side image-map
usemap	#mapname	Specifies an image as a client-side image-map

- **Example:**

```
<html>
    <body>
        <br>
        <img src="14.jpg"  width="400" height="200" align="right">
        <img  src="C:\\Users\\Admin\\Desktop\\Others\\13.jpg"     width="300"
        height="250" border="3" alt="Alternative Text">
    </body> </html>
```

Output:



Embedding media – audio, video

Embedding content involves integrating external elements, such as videos, audio, maps, or other HTML documents, seamlessly within a webpage.

Embedding Audio To embed audio content on a web page using HTML, you can use the <audio> tag, specifying the audio file's source using the "src" attribute.
<audio> -The <audio> tag is used to embed audio content on a web page.

Syntax:

```
<audio controls>
 <source src="audiofile.mp3" type="audio/mp3"> Your browser does not support the audio element.
</audio>
```

Attributes:

- **src : Specifies the URL of the audio file to be embedded.**
- **controls : Adds playback controls (play, pause, volume, etc.) to the audio player.**

Adding a Video Using the <video> Tag

```
<html>
<body>
   <video width="320" height="100" controls>
      <source src="movie.mp4" type="video/mp4">
      Your browser does not support the video tag.
   </video>
</body>
</html>
```

- The <video> tag is used to embed a video in the HTML page.
- The controls attribute adds play, pause, and volume control buttons.
- The <source> tag specifies the video file and format. If the video is not supported, the text inside the <video> tag will be displayed.

Canvas and SVG basics.

Both **<canvas>** and **<svg>** are used to create **graphics, drawings, shapes, and diagrams on a webpage**.

Using the <canvas> Element for Drawing

1. <canvas> — Drawing Area

Think of Canvas as a **blank drawing board**.

You use **JavaScript** to draw on it.

Basic syntax:

```
<canvas id="myCanvas" width="400" height="200"
    style="border:1px solid black;">
</canvas>
```

Here:

- `canvas` → creates the drawing area
- `id="myCanvas"` → gives the canvas a name
- `width="400"` → width of canvas
- `height="200"` → height of canvas
- `border` → makes the canvas visible

2. `<svg>` — Scalable Vector Graphics

SVG means **Scalable Vector Graphics**.

Think of SVG as **drawing using HTML tags**.

Unlike Canvas, SVG shapes are created using elements such as:

- `<rect>` → rectangle
- `<circle>` → circle
- `<line>` → line
- `<ellipse>` → ellipse
- `<polygon>` → polygon

```
<!DOCTYPE html>
<html>
<head>
    <title>SVG Example</title>
</head>

<body>
<h2>SVG Example</h2>

<svg width="400" height="200"
     style="border:1px solid black;">
    <rect x="50" y="50"
          width="150" height="80"
          fill="blue">
```

- </rect>
-
- </svg>
-
- </body>
- </html>
-
- x →
- 0 50 100 150 200
- ┌──────────────────────────────────────┐
- │ ┌──────────────────┐ │
- │ │ Rectangle │ │
- │ │ │ │
- │ └──────────────────┘ │
- └──────────────────────────────────────┘
- ↓
- Y
- x="50" → Start the rectangle 50 pixels from the left
- y="50" → Start the rectangle 50 pixels from the top
- width="150" → Rectangle is 150 pixels wide
- height="80" → Rectangle is 80 pixels high
- Responsive Web Design - The Viewport
- Setting The Viewport
- The viewport is the user's visible area of a web page.

- The viewport varies with the device (will be a lot smaller on a mobile phone than on a computer screen).
- You should include the following <meta> element in the <head> section of all your web pages:
- <meta name="viewport" content="width=device-width, initial-scale=1.0">
- This gives the browser instructions on how to control the page's dimensions and scaling.
- The width=device-width part sets the width of the page to follow the screen-width of the device (which will vary depending on the device).
- The initial-scale=1.0 part sets the initial zoom level when the page is first loaded by the browser.
- Size Content to The Viewport
- Users are used to scroll websites vertically on both desktop and mobile devices - but not horizontally!
- So, if the user is forced to scroll horizontally, or zoom out, to see the whole web page it results in a poor user experience.
- Some additional rules to follow:
- **Do NOT use large fixed-width elements -** For example, if an image has a width wider than the viewport, it causes the viewport to scroll horizontally
- **Use CSS media queries to apply different styling for small and large screens** - Setting large absolute CSS widths for page elements will cause the elements to be too wide for smaller devices. Instead, consider using relative width values, such as width: 100%.
- **ARIA = Accessible Rich Internet Applications**
- ARIA is used to make a webpage **more accessible to people with disabilities**, especially users who use **screen readers**. It adds extra information about HTML elements so assistive technologies can understand them

Assistive Technology	Helps whom?	What it does
Screen Reader	Visually impaired users	Reads webpage content aloud

With ARIA:

```
<div
   role="progressbar"
   aria-valuenow="75"
   aria-valuemin="0"
   aria-valuemax="100">
</div>
```

Here:

role="progressbar" → tells what the element is

aria-valuemin="0" → minimum = 0

aria-valuemax="100" → maximum = 100

aria-valuenow="75" → current progress = 75%

ARIA adds accessibility information to HTML elements so that assistive technologies such as screen readers can understand and interact with web content.

- Accessibility Standards – Alt Text & Best Practices
- **Web accessibility** means designing a website so that **everyone, including people with disabilities, can use and understand it.**
- 1. What is Alt Text?

- **Alt text = Alternative Text**
- It is a short description of an image written using the alt attribute.
- <img src="college.jpg" alt="DY Patil college building">
- If the image cannot load, the alt text can be displayed.
- More importantly, a screen reader can read the alt text to a visually impaired user.
- <img src="college.jpg" alt="image">(Not Useful)
- <img src="college.jpg" alt="Students attending a computer science lecture">(Useful)
- Important Accessibility Best Practices

• Practice	• Example
• **Use meaningful alt text**	• alt="Students in computer lab"
• **Use semantic HTML**	• <nav>, <header>, <main>, <button>
• **Use proper headings**	• <h1>, <h2>, <h3>
• **Provide labels for forms**	• <label for="email">Email</label>
• **Ensure keyboard accessibility**	• Users should be able to use Tab/Enter

• Practice	• Example
• **Use sufficient color contrast**	• Text should be clearly visible
• **Provide captions**	• Useful for videos
• **Don't rely only on color**	• Don't say only "click the red button"
• **Make links descriptive**	• "Download timetable" instead of "Click here"
• **Use ARIA when necessary**	• Add accessibility information when native HTML isn't enough

Accessibility Standards

- A major international standard is **WCAG — Web Content Accessibility Guidelines**, developed by the **W3C**.
- WCAG is commonly explained through four principles:
- **P-Perceivable**
  Information should be available to the user's senses.
- **O-Operable**
  Users should be able to operate the website, including using a keyboard.
- **U-Understandable**
  Content and navigation should be easy to understand.

- **R-Robust**

  Content should work with different browsers and assistive technologies.